

CS & IT ENGINEERING



Operating System

Virtual Memory

Lecture – 6

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Recap of Previous Lecture



Topic

Frame Allocation

Topic

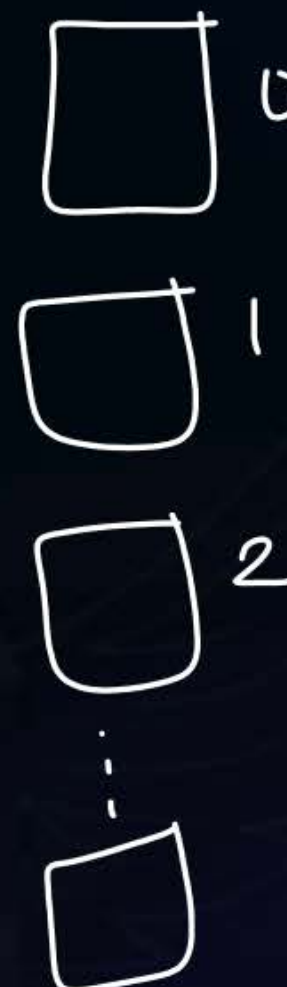
Thrashing

Topic

Multilevel Paging



Process



Topics to be Covered



Topic

Multilevel Paging

Topic

Average Memory Access in Multilevel Paging

Topic

Inverted & Hashed Page Table

in multilevel paging, levels are increased till the outermost page table can fit into a single page



Topic : Question

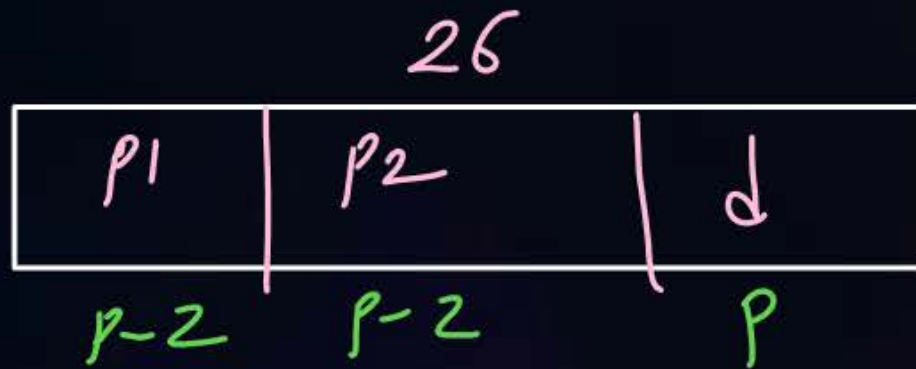
VA = 26 bits

Page table entry size = 4 bytes

2-level paging

Outer page table fits into a page exactly

Page size = 1 Kbytes?



$$2(p-2) + p = 26$$

$$3p = 30$$
$$p = 10$$

max p.T. entries

$$\text{per page} = \frac{2^p \text{ Bytes}}{4 \text{ Bytes}}$$

$$= 2^{p-2}$$

$$\text{Page size} = 2^{10} \text{ B}$$
$$= 1 \text{ KB}$$



Topic : Question

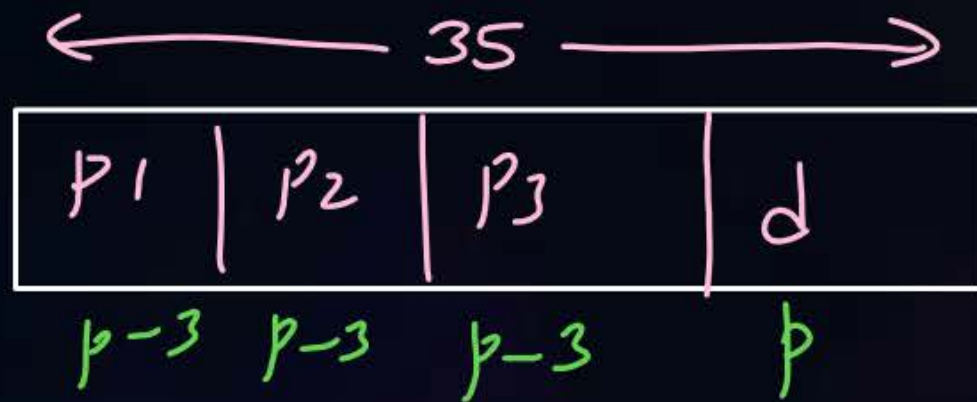
VA = 35 bits

Page table entry size = 8 bytes

3-level paging

Outer page table fits into a page exactly

Page size = 2 Kbytes?



$$3(p-3) + p = 35$$

$$4p = 44$$

$$p = 11$$

$$\text{page size} = 2^{11} \text{ B} = 2 \text{ KB}$$

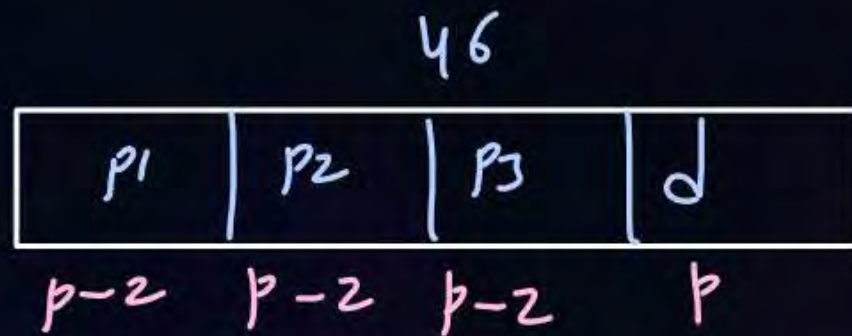


Topic : Question

Ans = 8

#Q. A computer uses 46-bit virtual address, 32-bit physical address, and a three-level paged page table organization. The page table base register stores the base address of the first-level table (T1), which occupies exactly one page. Each entry of T1 stores the base address of a page of the second-level table (T2). Each entry of T2 stores the base address of a page of the third-level table (T3). Each entry of T3 stores a page table entry (PTE). The PTE is 32 bits in size. What is the size of a page in KB in this computer?

↓
4 bytes



$$3(p-2) + p = 46$$

$$4p = 52$$

$$p = 13$$

$$\text{page size} = 2^{13} = 8 \text{ KB}$$



Topic : Question

Size of page tables across all levels?

← 19 →		
P1	P2	D
1	8	10

assumed page size here = $2^{10} \text{ B} = 1 \text{ KB}$

p.T. entry size = 4 Bytes

No. of pages required to store outermost page table = 1

|| Inner p.T. = $2^{P_1} = 2^1 = 2$

Total no. of pages = 3

p.T. size across all levels = $3 * \text{page size} = 3 * 1 \text{ KB}$
 $= 3 \text{ KB}$





Topic : Question

Size of page tables across all levels?

P1	P2	D
2	9	12

→ Page size = $2^{12} \text{ B} = 4 \text{ KB}$

$$\begin{array}{rcl} \text{no. of pages for outer P.T.} & = & 1 \\ \text{---||---Inner---||---} & = & 2^2 = 4 \\ \hline \text{Total} & = & 5 \end{array}$$

$$\text{Total size} = 5 * 4 \text{ KB} = 20 \text{ KB}$$



Topic : Question

Size of page tables across all levels?

P1	P2	P3	D
1	8	8	10

$$\text{Page size} = 2^{10} \text{B} = 1 \text{KB}$$

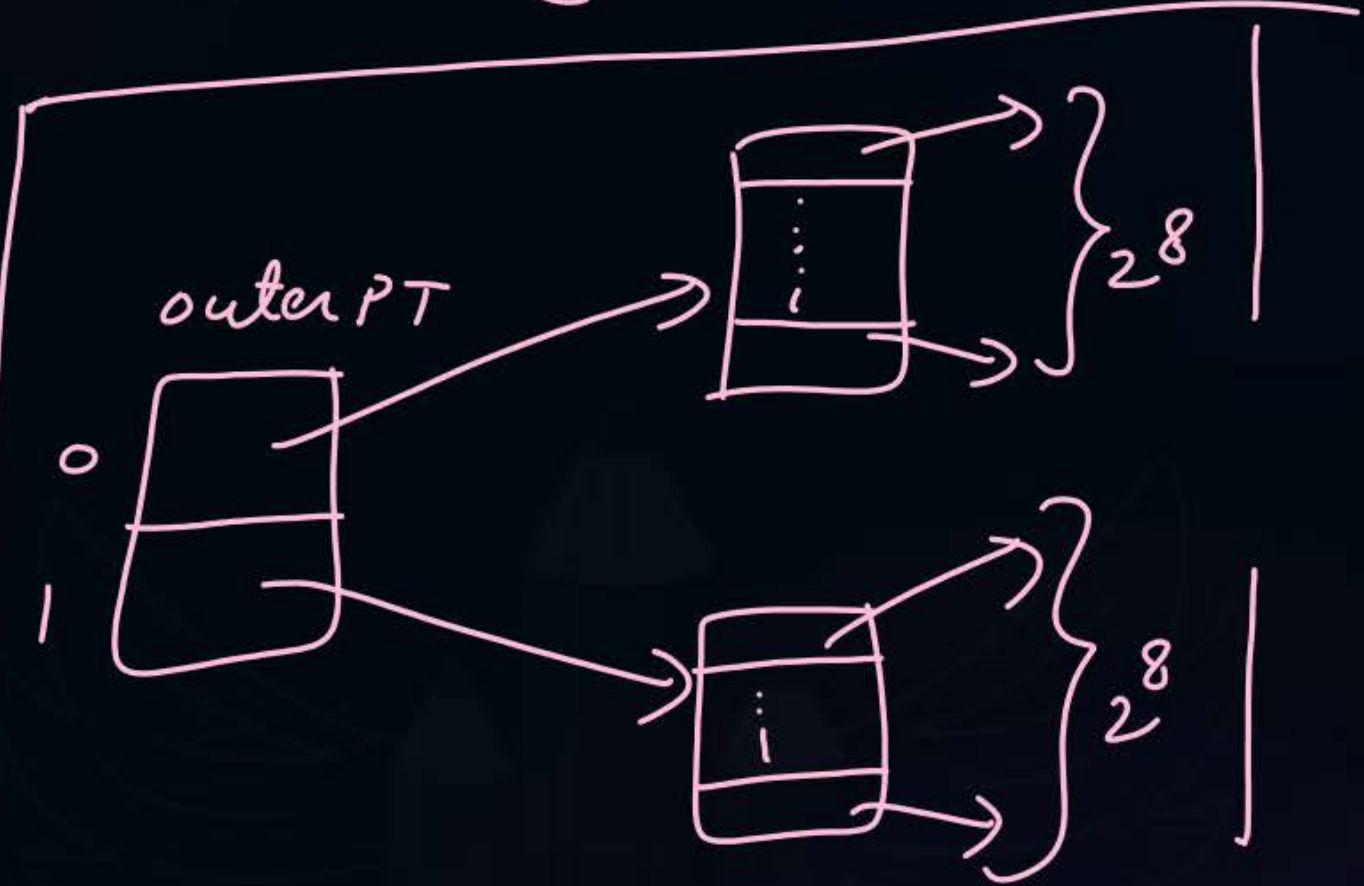
No. of pages for outer P.T. = 1

——||—— for middle P.T. = $2^1 = 2$

——||—— for inner P.T. = $2^{P_1} * 2^{P_2} = 2^1 * 2^8$
 $= 2^9 = 512$

Total = 515

p.t. size across all levels = $515 * 1 \text{KB}$
 $= 515 \text{KB}$





no. of pages for outer P.T. = 1

—— 11 ——— middle P.T. = $2^3 = 8$

—— 11 ——— inner P.T. = $2^3 * 2^8 = 2^{11} = 2048$

Total = 2057

Total P.T. across all levels = $2057 * 2KB$
 $= 4114 KB$

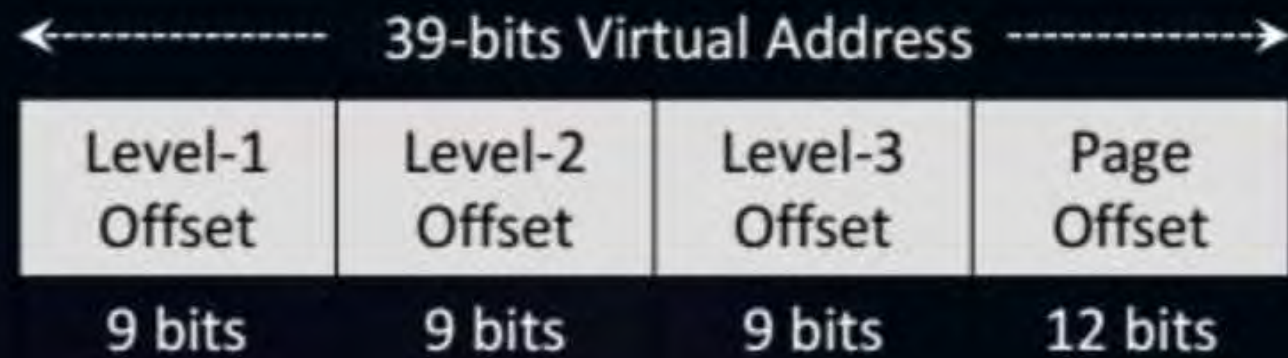
32-bit OS $\Rightarrow$ max allowed v.A. for any process = 32-bits

——||—— process size = 2^{32} B = 4GB



Topic : Question

#Q. Size of page Consider a three-level page table to translate a 39-bit virtual address to a physical address as shown below:



The page size is 4 KB = (1KB = 2^{10} bytes) and page table entry size at every level is 8 bytes. A process P is currently using 2 GB ($1 \text{ GB} = 2^{30}$ bytes) virtual memory which OS mapped to 2 GB of physical memory. The minimum amount of memory required for the page table of P across all levels is 4168 KB across all levels?

Handwritten note: 2^{31} B with an arrow pointing to the circled '2 GB'.

$$31$$

P_1	P_2	P_3	d
1	9	9	12

no. of pages for outer p.t. = 1

—— || —— middle p.t. = $2^1 = 2$

—— || —— inner p.t. = $2 * 2^9 = 1024$

—————
Total = 1027

Total space = $1027 * 4KB$
= 4108 KB



Topic : Question

Home-work



#Q. Consider a 32-bit system with 4 KB page size and page table entries of size 4 bytes each. Assume $1 \text{ KB} = 2^{10}$ bytes. The OS uses a 2-level page table for memory management, with the page table containing an outer page directory and an inner page table. The OS allocates a page for the outer page directory upon process creation. The OS uses demand paging when allocating memory for the inner page table, i.e. a page of the inner page table is allocated only if it contains at least one valid page table entry.

An active process in this system accesses 2000 unique pages during its execution, and none of the pages are swapped out to disk. After it completes the page accesses, let X denote the minimum and Y denote the maximum number of pages across the two levels of the page table of the process.

The value of $X + Y$ is _____?



Topic : EMAT With Multilevel Paging

multiple mm accesses are needed for P.T. access

2-level paging:-

$$EMAT = H * (t_{TLB} + t_{mm}) + (1-H) \left(t_{TLB} + \frac{2t_{mm}}{\uparrow} + t_{mm} \right)$$

for 2 level paging



Topic : Question

H.W. Questⁿ

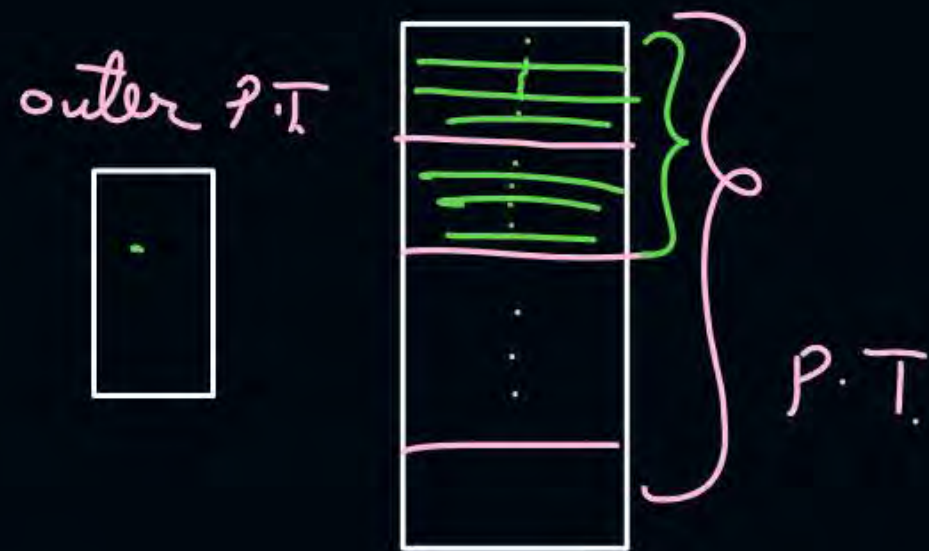
- #Q. A processor uses 2-level page tables for virtual to physical address translation. Page tables for both levels are stored in the main memory. Virtual and physical addresses are both 32 bits wide. The memory is byte addressable. For virtual to physical address translation, the 10 most significant bits of the virtual address are used as index into the first level page table while the next 10 bits are used as index into the second level page table. The 12 least significant bits of the virtual address are used as offset within the page. Assume that the page table entries in both levels of page tables are 4 bytes wide. Further, the processor has a translation look-aside buffer (TLB), with a hit rate of 96%. The TLB caches recently used virtual page numbers and the corresponding physical page numbers. The processor also has a physically addressed cache with a hit rate of 90%. Main memory access time is 10 ns, cache access time is 1 ns, and TLB access time is also 1 ns. Assuming that no page faults occur, the average time taken to access a virtual address is approximately (to the nearest 0.5 ns)?

Single-level paging



entire P.T. must be stored
in mm always

Multilevel paging



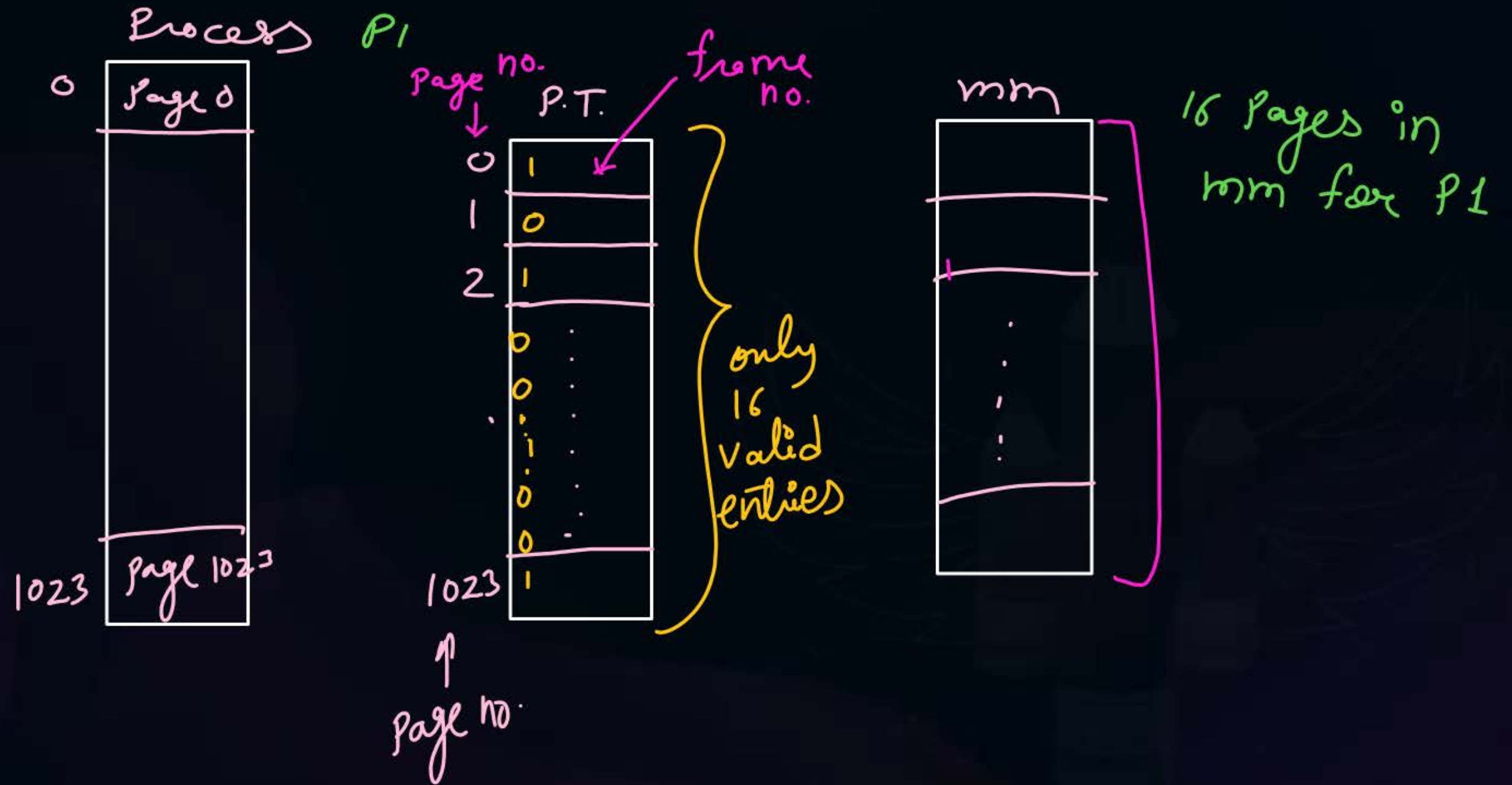
Beneficial when only a few pages of
P.T. are frequently needed hence remaining
pages of P.T. can be kept on disk to
save lot of space of m.m.

Ques) Home work

How page table size is dependent on page size? (single level paging)



Page table is too large and so many page table entries are invalid.





Inverted P.T.

no. of entries
= no. of frames

only single P.T. maintained
for all processes

Frame no.

Page 2 of Process P1	0
Page 15 of Process P3	1



Topic : Inverted Page Table

Process

0	
1	
2	
3	
4	
5	
6	
7	

Page Table

0	
1	
2	1
3	
4	3
5	
6	0
7	2

Inverted Page Table

P1	6	0
P1	2	1
P1	7	
P1	4	

Physical Memory

0	Page 6
1	Page 2
2	Page 7
3	Page 4



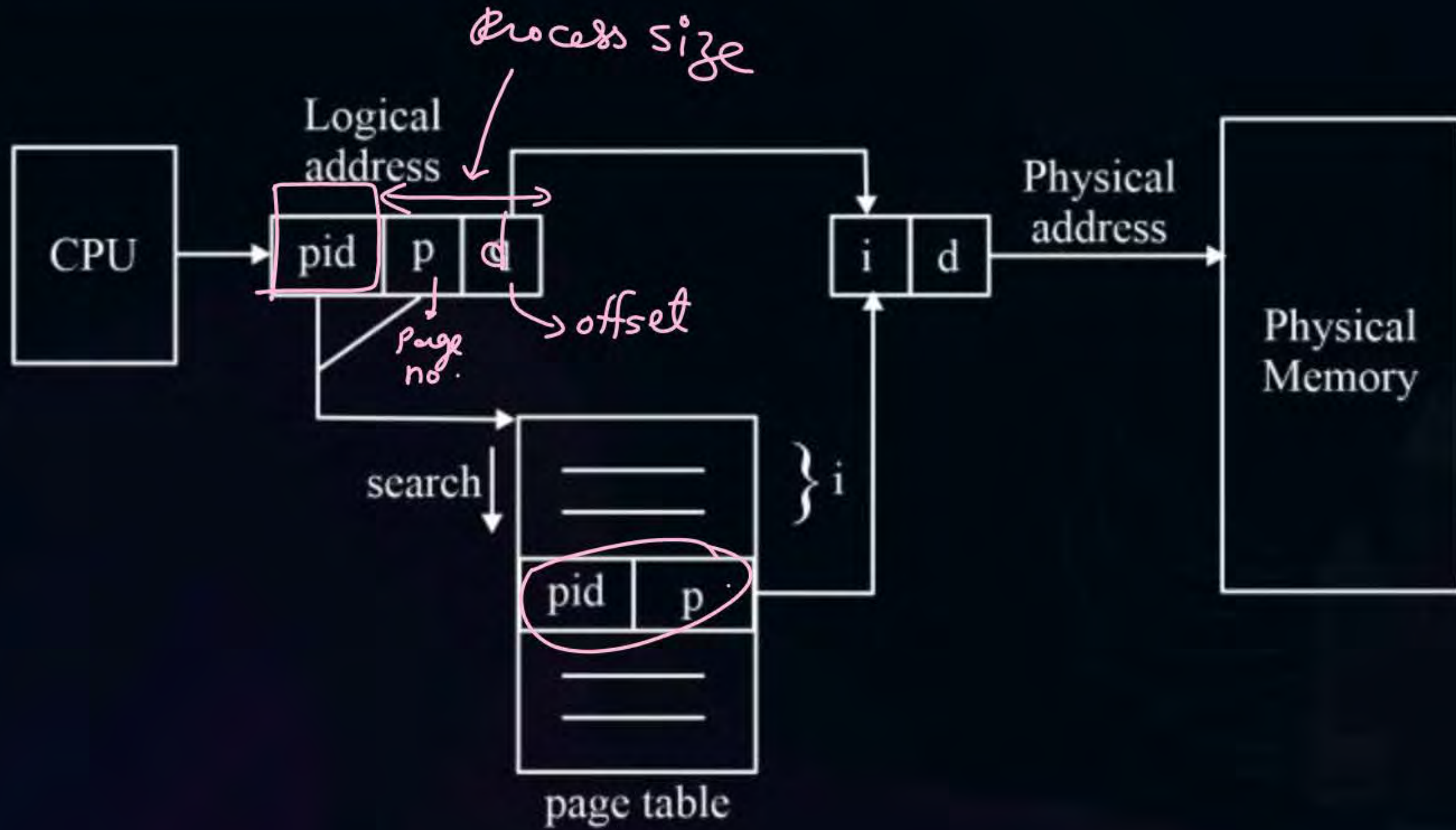
Topic : Inverted Page Table

Each entry in the page table contains the following fields:

1. Page number
2. Process id
3. Control bits
4. Chained pointer



Topic : Inverted Page Table





Topic : Inverted Page Table

Advantages & Disadvantages:

1. Reduced memory space
2. Longer lookup time
3. Difficult shared memory implementation



2 mins Summary

Topic

Multilevel Paging

Topic

Average Memory Access in Multilevel Paging

Topic

Inverted & Hashed Page Table



Happy Learning

THANK - YOU